

UCab — Full-Stack MERN Cab Booking Platform

Solo Developer / Team Member: Shaik Sumiya Zainab | Project ID: N/A (Solo Track)

Phase 3: Project Design Phase — Problem-Solution Fit & Value Proposition

Project Name: Cab Booking (‘UCab’)
Project ID: ‘N/A (Solo Track Submission)’
Team Member / Solo Developer: Shaik Sumiya Zainab

1. Value Proposition Canvas Overview

To ensure complete Problem-Solution Fit, UCab was structured around the exact customer jobs, pains, and gains identified during Phase 1 research. Every technical feature built into the application directly resolves a specific friction point for urban commuters and fleet dispatchers.

```
graph LR
  subgraph Customer Profile [Urban Commuter Profile]
    Jobs[Customer Jobs: Quick commute, tax receipts, safe transit]
    Pains[Pains: Hidden pricing surges, unclear driver assignment, app crashes]
    Gains[Gains: Transparent fares, in-ride comfort, printable invoices]
  end
  subgraph Value Map [UCab MERN Value Map]
    Products[Products & Services: Multi-Category Fleet, Live Dispatch Tracker]
    Relievers[Pain Relievers: Deterministic Fare Calculator, Zero-Config Engine]
    Creators[Gain Creators: In-Ride Refreshments Package, QR Corporate Receipts]
  end
  Relievers --> Pains
  Creators --> Gains
  Products --> Jobs
```

2. Problem-Solution Alignment Matrix

Identified Customer / Admin Pain Point	UCab Value Proposition & Feature Solution	Technical Mechanism
Opaque, fluctuating cab fares with hidden surge charges.	**Deterministic & Transparent Pricing Engine:** Users see explicit base fares and exact per-km rates for every cab class before booking. Real-time calculation formula inside ‘BookCab.jsx’ and ‘bookingController.js’.	
Impersonal travel experience with no hospitality options.	**In-Ride Customization & Charity Suite:** Commuters can pre-order chilled water & snacks (+ ₹50) or donate to driver healthcare (+ ₹20). State-driven checkout add-on checkboxes stored directly in ‘Booking’ schema.	
Uncertainty about driver dispatch and vehicle arrival.	**Interactive Live Dispatch Status Bar:** Visual step-by-step progress tracking with exact driver name (‘Vikram Sharma’) and plate number. Dynamic state mapping (‘Pending’ ! ‘Accepted’ ! ‘Started’ ! ‘Completed’) in ‘UserHome.jsx’.	
Difficulty generating verified receipts for company tax claims.	**Instant Corporate Thermal Invoice Modal:** One-click PDF receipt generation complete with an embedded verification QR code. ‘ReceiptModal.jsx’ utilizing clean table formatting and dynamically generated QR strings.	
Database installation crashes during university/presentation grading.	**Zero-Config Dual Database Storage Engine:** Guarantees 100% demo uptime whether MongoDB is online or offline. Automatic detection and atomic fallback (‘server/db/store.js’) saving to ‘data.json’.	